

10 AI Engineering Terms Every Developer Should Know

Master the language behind modern AI applications.



Understand
core concepts



Build
better systems



Ship
smarter apps



Scale
with confidence



Stay ahead
in the AI era

1. Tokens



A token can be a whole word, part of a word, a space, or punctuation.

Tokens are the **smallest units of text** an LLM processes.



EXAMPLE

“ChatGPT is amazing”



“ChatGPT” , “ is” , “ amazing”



Illustrative example only.
Actual tokenization varies by model.



WHY IT MATTERS



Context windows are measured in tokens.



Token usage affects cost.



More tokens = more processing.

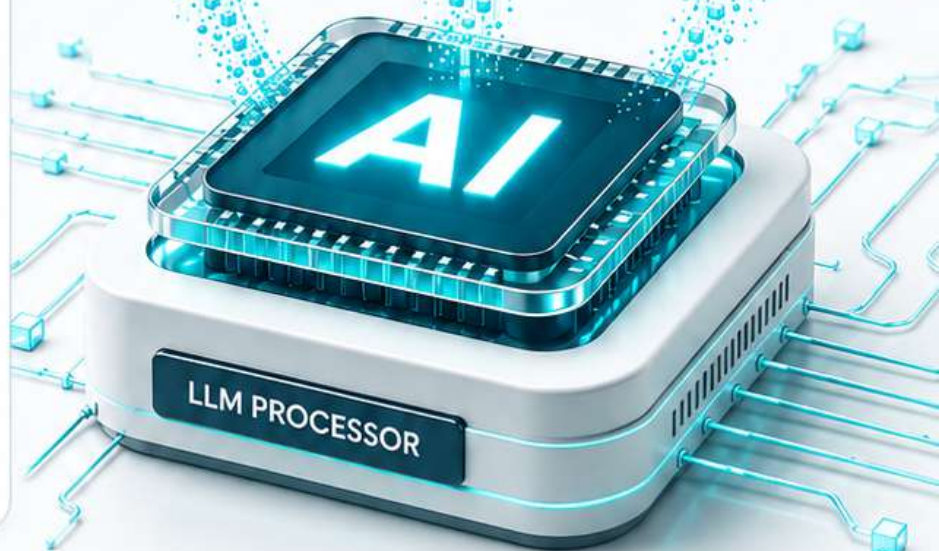


TIP: Shorter prompts use fewer tokens and are more efficient.

INPUT TEXT

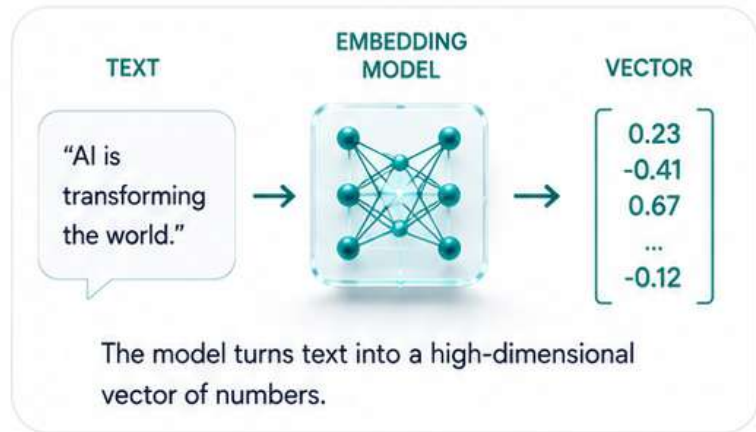
“ChatGPT is amazing”

TOKENIZATION (Illustrative)



2. Embeddings

Embeddings convert text into **numerical vectors** that capture **semantic meaning**.



WHY IT MATTERS



Semantic Search

Finds meaning, not just keywords.



Recommendations

Understands user intent and context.



Similarity Search

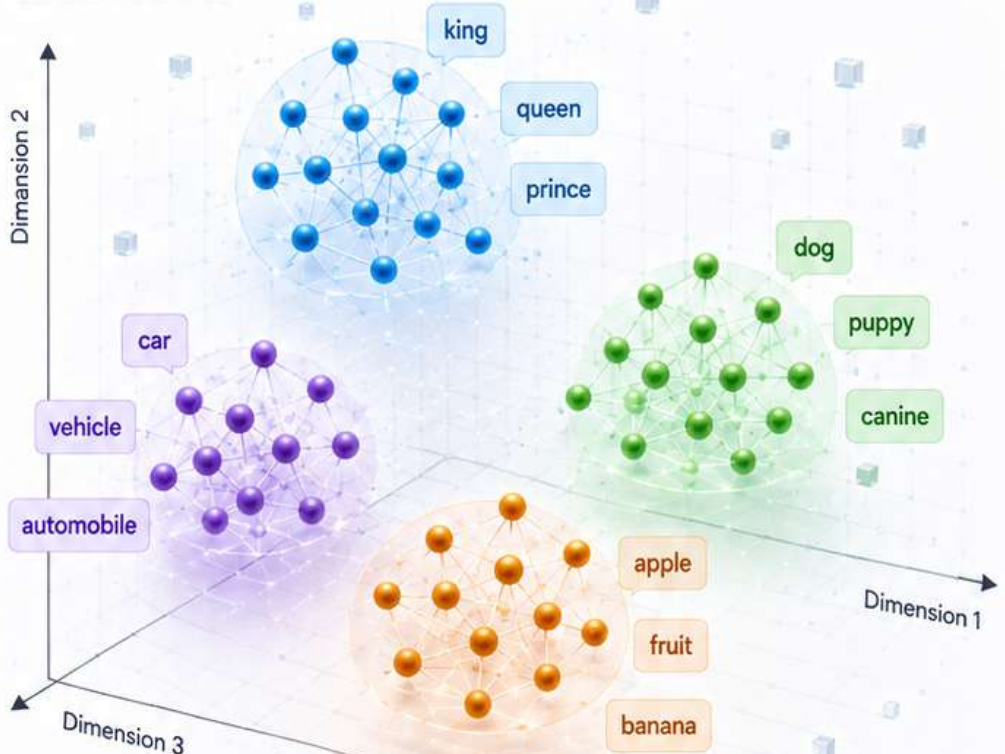
Finds the closest match in meaning.



AI Memory

Stores and recalls information intelligently.

SEMANTIC SPACE

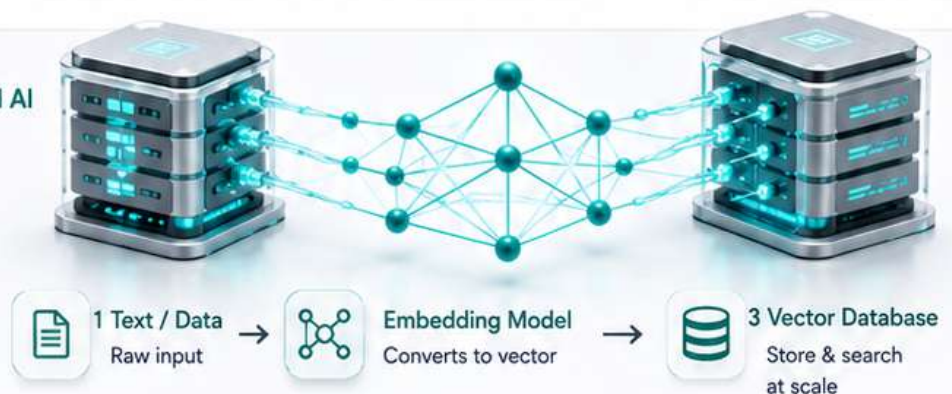


Similar meanings are closer in vector space.



EMBEDDINGS POWER MODERN AI

Embeddings map meaning to numbers so machines can **understand relationships**, make predictions, and deliver smarter experiences.



KEY TAKEAWAY

Embeddings capture the **meaning** behind text — not just the words.

3. Vector Database

A specialized database designed to **store and search embeddings** efficiently.

WHAT IS IT?

Text / Data



Embedding
(Vector)

[0.21, -0.17, 0.63, ...]
[0.45, 0.12, -0.33, ...]
[0.33, -0.78, 0.11, ...]
...



Vector
Database



★ WHY IT MATTERS



Instead of keyword matching, it finds information based on **meaning**.



Understands context and semantics



Finds the most relevant results



Works with unstructured data at scale



Powers modern AI applications (RAG, semantic search, recommendations)



HOW IT WORKS



Question

User asks something



Embedding

Convert question to vector



Vector Database

Search for similar vectors



Relevant Results

Return the most relevant information

4. Context Window

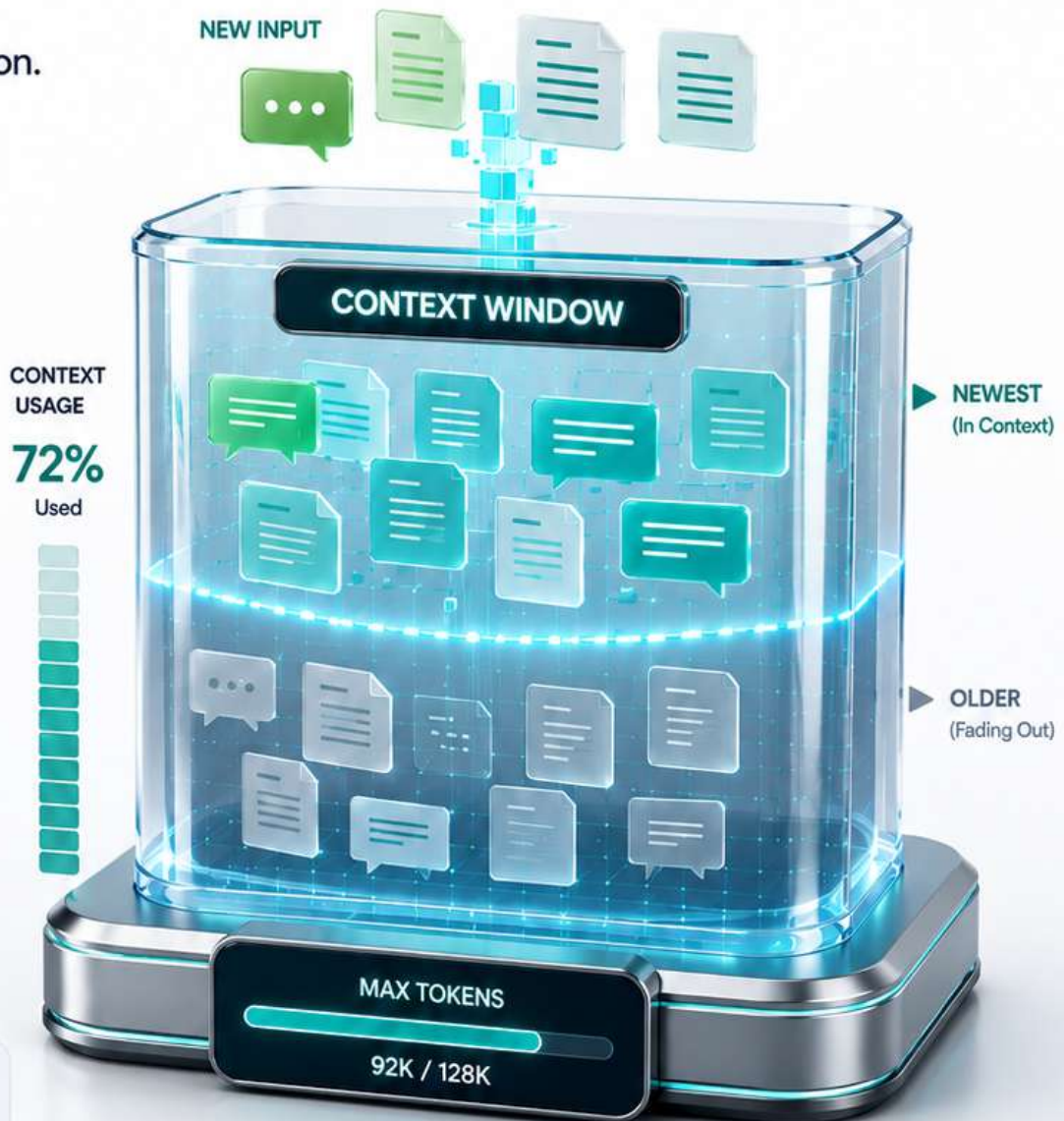
The **maximum amount** of information an LLM can process in one conversation.



★ WHY IT MATTERS

- Long chats**
Keep more history in context.
- Large documents**
Process more content at once.
- Better reasoning**
More context leads to smarter answers.
- Better consistency**
Maintains coherence across longer conversations.

Tip: Bigger context windows unlock more powerful AI applications.



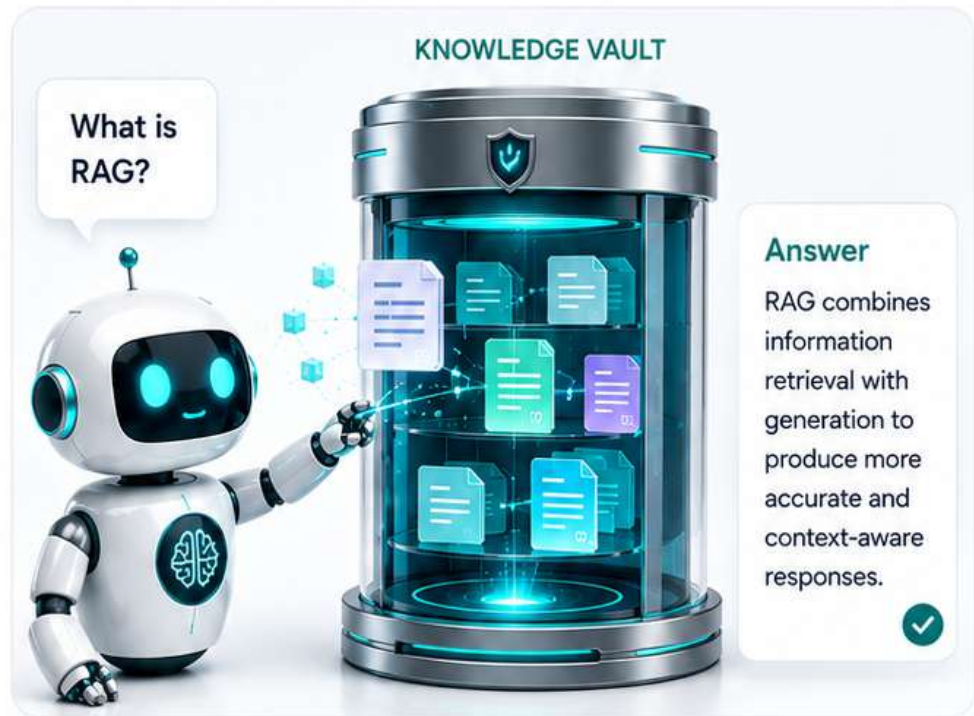
HOW CONTEXT WORKS



5. RAG

Retrieval-Augmented Generation

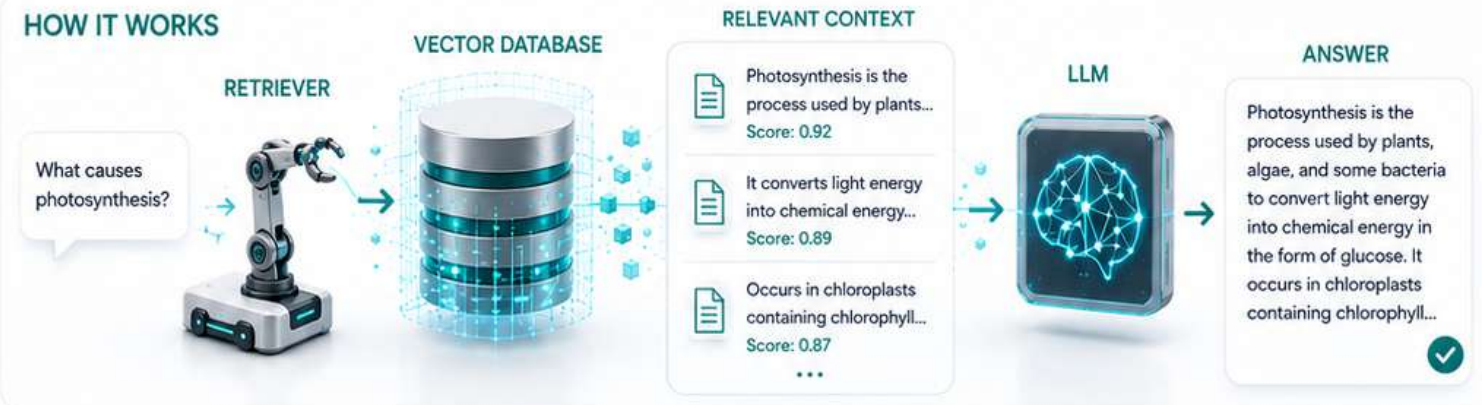
Retrieves **relevant information** **before** asking the LLM to generate a response.



THE RAG PIPELINE



HOW IT WORKS



WHY IT MATTERS

Reduces hallucinations and improves accuracy.



More accurate answers



Fewer hallucinations



Up-to-date information

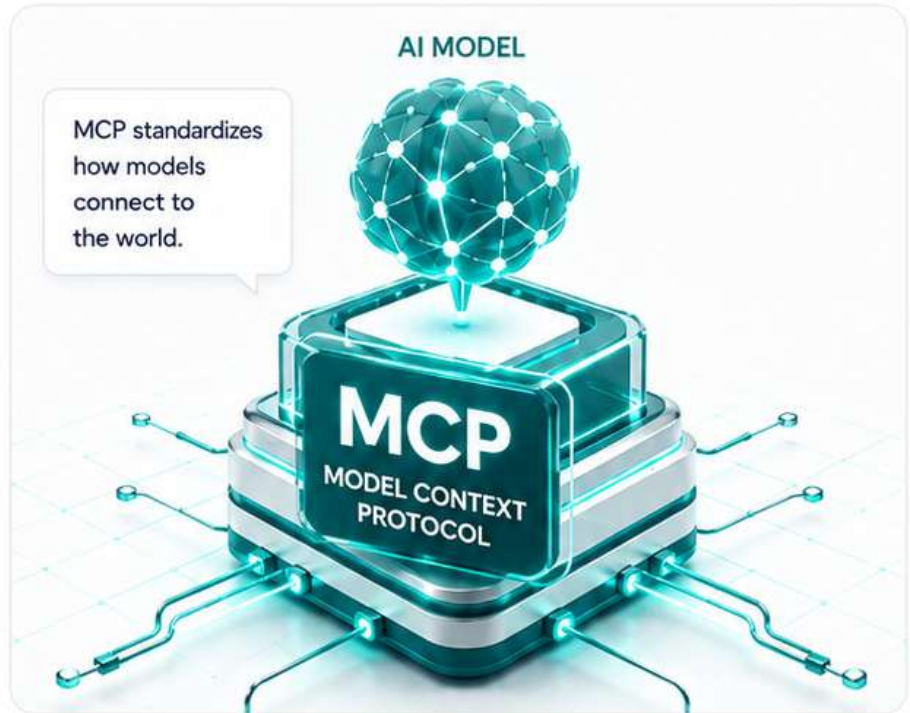


Better trust & reliability

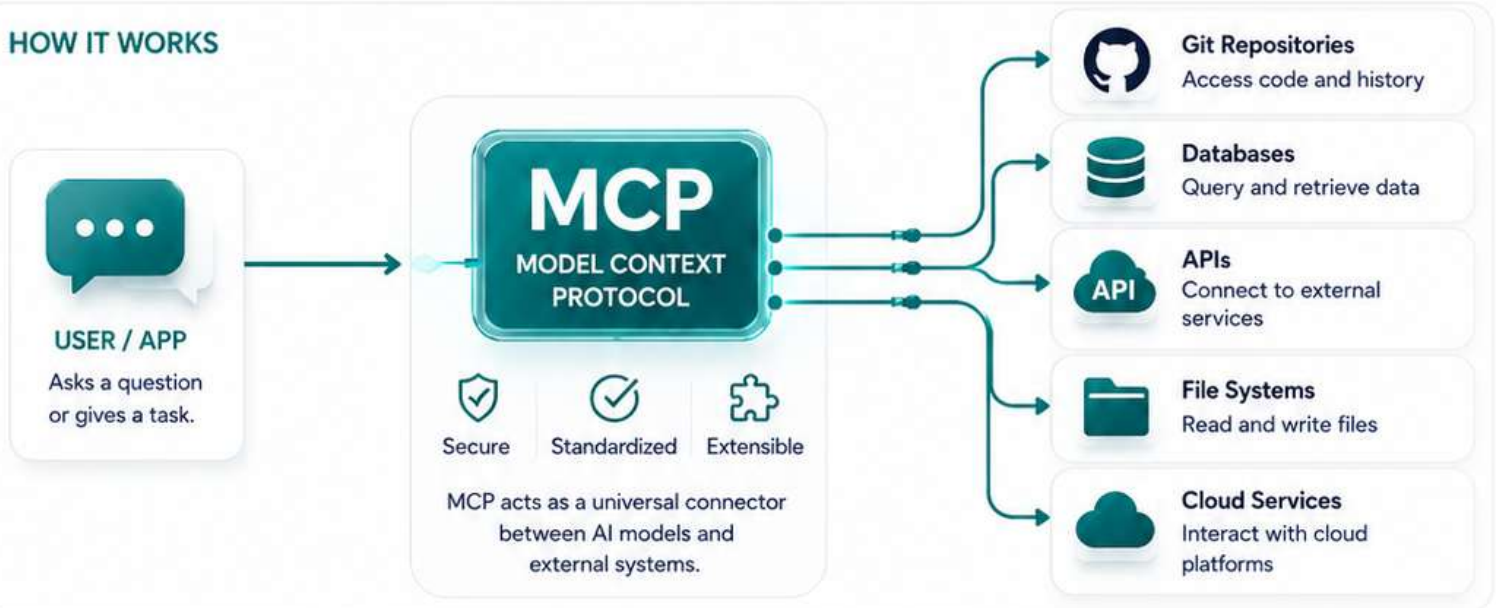
6. MCP

Model Context Protocol

A standardized way for AI models to **communicate** with **external tools, APIs,** and **data sources.**



HOW IT WORKS



WHY IT MATTERS

One protocol.
Many integrations.

MCP eliminates custom integrations for every tool and enables AI models to securely and efficiently interact with thousands of systems.

KEY BENEFITS



Interoperability

Works across different tools and platforms.



Efficiency

Faster development with reusable integrations.



Security

Permissioned access with built-in safeguards.



Scalability

Easily connect to new tools and data sources.

EXAMPLE USE CASES



AI coding assistant accessing your repo



AI agent querying internal databases



AI tool analyzing logs and metrics



AI chatbot retrieving knowledge base data



AI workflow triggering external APIs

7. AI Agent

An AI system that can **reason, plan, use tools,** and **complete tasks autonomously.**

★ WHY IT MATTERS



AI Agents automate complex workflows beyond simple chat.



THE AI AGENT WORKFLOW



WHAT AI AGENTS CAN DO



EXAMPLE AI AGENT IN ACTION



User: Generate a weekly sales report and email it to the team.



Agent: Understood. I'll analyze sales data, create the report, and send it via email.



Analyze Data



Generate Report



Email Team



Update CRM



Agent: Done! The report has been emailed and CRM updated.



AI Agents are the bridge between intelligence and real-world action.

Autonomous

Goal-Oriented

Tool-Powered

Adaptive

8. Tool Calling

9. Function Calling



8. Tool Calling

Allows AI to interact with **external tools** like databases, APIs, search engines, or file systems.

EXAMPLES



Web Search
Search the web for information.



APIs
Get real-time data from services.



Databases
Read or write data securely.



File Systems
Read, write or manage files and documents.



Search Engines
Retrieve relevant results.

You ask.
AI decides
what to do.



AI acts using
tools or functions.



9. Function Calling

Lets the LLM invoke **predefined application functions** with structured arguments.

EXAMPLES



get_weather
{ "location": "New York" }



create_invoice
{ "customer_id": 123, "amount": 499 }



send_email
{ "to": "user@example.com",
"subject": "Report" }



get_user_profile
{ "user_id": 42 }



calculate_tax
{ "amount": 1000, "country": "US" }

HOW IT WORKS



1. User Request
You ask a question or give a task.



2. AI Understands
LLM understands the intent.



3. Decide
Choose between using a tool or a function.



4. Execute
Call the tool or function with the right inputs.



5. Observe Result
Get the result from the external system.



6. Respond
Return the final answer to the user.



**Together, They Enable
AI to Perform Real-World Actions.**



Go beyond
simple chat



Interact with
real systems



Automate
complex tasks



Deliver real
business value

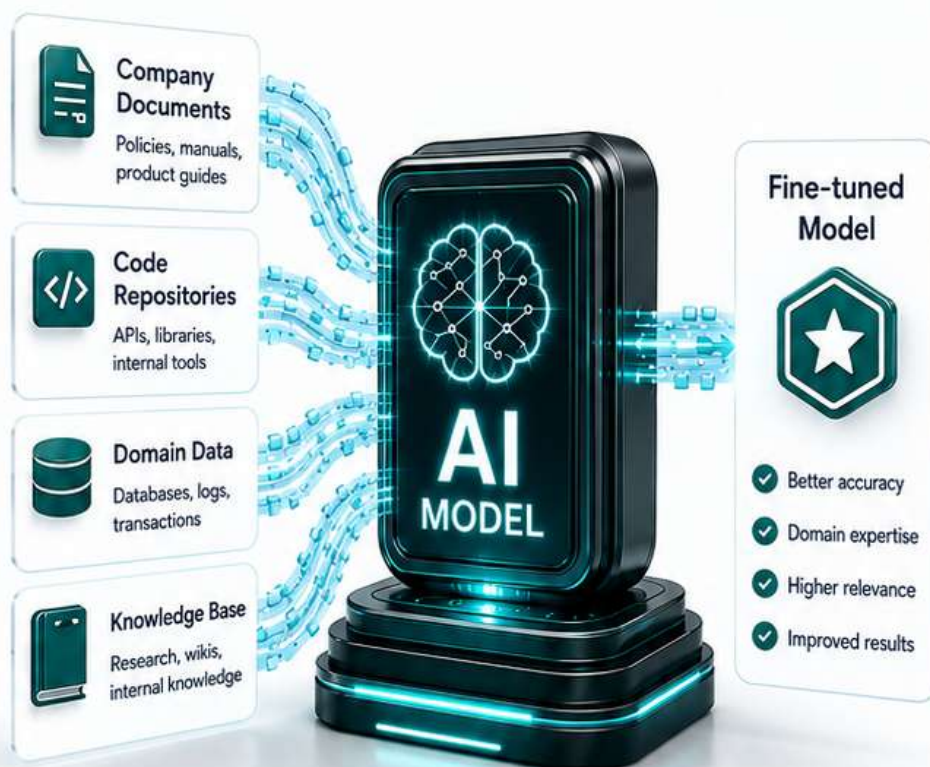
10. Fine-tuning

Training a pre-trained model further on **domain-specific data** to improve performance for specialized tasks.

WHEN TO USE FINE-TUNING

- ✓ Industry-specific knowledge
- ✓ Company terminology
- ✓ Custom writing style
- ✓ Specialized workflows

MAKE THE MODEL AN EXPERT IN YOUR DOMAIN

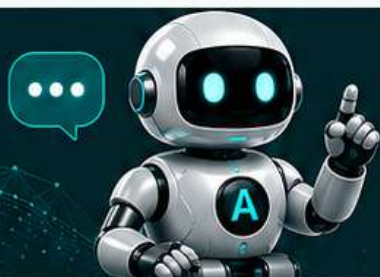


FINE-TUNING TRANSFORMS A GENERAL MODEL INTO A DOMAIN EXPERT



- Understands your domain
- Uses your terminology
- Follows your patterns
- Delivers expert-level results

THE COMPLETE AI ENGINEERING FLOW



Which AI engineering concept do you use **most often**?

Comment below

